LLM-BASED WHEEL HUMAN INTERACTION ROBOT FOR RECEPTIONIST AND HUMAN ASSISTANT

KIRAWUT CHALERMKITPAISAN
KRITTIN SAKHARIN
NATCHA RUNGRUANG
NATWASA MANOMAIWIBOON

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF
BACHELOR OF ENGINEERING IN ROBOTICS AND AI ENGINEERING
SCHOOL OF ENGINEERING
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KMITL-2025-EN-X-XXX-XXX

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Thesis	LLM-based Wheel Human Interaction Robot for Receptionist and Human Assistant
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Degree	Bachelor of Engineering
Program	Robotics and AI Engineering
Year	2025
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ABSTRACT IN ENGLISH

A good abstract writing should focus only on the key and interesting aspects of the research. It should emphasize the specific highlights of the research, with the research work being clear, concise, and meeting certain criteria for a good abstract. For example, the word count should be between 200-250 words, or approximately no more than half of an A4 page. Before writing the abstract, one should read and understand their own complete research work to identify various interesting points that can engage general readers, making them want to read the full research. There should be no interpretation or criticism using personal opinions. Complex or aggressive language should be avoided. Local jargon, unnecessary abbreviations, or symbols should not be used as they may lead to misunderstanding. There should be no numerical references, diagrams, tables, various statistical formulas, or equations in the abstract, unless necessary to display analytical results. The abstract may have multiple paragraphs to enhance reader comprehension in each section. Avoid referencing the research work of others in the abstract. The structure of the abstract should also be carefully considered, including font type, font size, page margin, printing standards, and an acceptable format. Proofreading the writing by reading it multiple times will improve the quality of the abstract, making it acceptable for research presentations.

ACKNOWLEDGEMENT

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Name Surname

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LIST OF ABBREVIATIONS AND SYMBOLS

C	Specific heat
P	Pressure
V	Volume

CHAPTER 1

INTRODUCTION

1.1 Statement and Significance of the Problems

In modern educational institutions, students often require quick access to information and assistance in navigating campus environments. However, traditional means of support, such as static information boards or human guides, may not always be available or efficient. With advancements in artificial intelligence and robotics, interactive robotic assistants have the potential to bridge this gap by providing real-time navigation support, answering queries, and improving overall student engagement. The proposed robotic assistant utilizes Simultaneous Localization and Mapping (SLAM) for autonomous navigation, voice recognition for user interaction, and Retrieval-Augmented Generation (RAG) to generate precise responses based on curriculum data. By integrating these technologies, the system aims to offer a seamless and intelligent support mechanism for students within the E12 Building at KMITL.

1.2 Goal and Objective of the Study

- **Achieve Autonomous Indoor Navigation:** Design and integrate a mobile robotic platform with LiDAR sensors for real-time obstacle detection and SLAM-based navigation within the E12 Building.
- **Integrate an LLM for Natural Conversation:** Develop a conversational agent using a Large Language Model to enable natural, context-aware interactions with students.
- **Implement RAG for Curriculum-Based Responses:** Utilize Retrieval-Augmented Generation to enhance the LLM's ability to provide precise, curriculum-aligned answers.
- **Develop a Robust Data Management System:** Use MySQL and Milvus to store and manage interaction data, curriculum information, and chat history.
- **Enable Voice Interaction:** Implement Speech-to-Text (STT) and Text-to-Speech (TTS) pipelines for hands-free interaction.
- **Implement a User-Friendly Interface:** Design an accessible interface for students with varying technological proficiency.
- **Optimize for Real-Time Performance:** Ensure seamless processing of interactions, navigation, and response generation.

1.3 Hypothesis

1.4 Scope of the Study

This study focuses on the development and evaluation of the robotic assistant within a defined operational environment. The robot is deployed exclusively on the 12th floor of the E12 Building, which serves

as the primary testing and demonstration area. All navigation mapping, waypoint configuration, and environmental testing are conducted within this space, and the system is not designed or evaluated for operation beyond this boundary.

The target users of the system are students of the Robotics and Artificial Intelligence (RAI) programme. Interactions are therefore tailored to the needs and contexts relevant to this user group within the building environment. The robot is designed to support core interaction scenarios: providing navigation guidance to designated locations within the floor, and engaging in general conversation through the LLM-powered dialogue system.

1.5 Process of the Study

1.5.1 Routine

1. State 1: Standby Mode

The robot positions itself at the starting station and waits for a manual command or wakeword to transition to State 2.

2. State 2: Walk Around

The robot navigates autonomously using SLAM. If the battery is running low, it returns to State 1. If a person is detected, it transitions to State 3.

3. State 3: Interact with Speaker

The robot engages in conversation using NLP and LLM. Once the conversation ends, it returns to State 2.

1.5.2 Hardware

LiDAR for obstacle detection, MongoDB for robust data storage, and gesture 456 recognition systems for intuitive interactions. 345

Following the mechanical design, the hardware system is developed by selecting appropriate electronic components, including the main controller, sensors for perception, actuators for movement, and the power management system. The hardware architecture is designed to ensure efficient communication between subsystems and to support the required functionalities of the robot. This stage establishes the foundation for integrating the hardware with software components in subsequent steps.

1.5.3 Framework

Retrieval-Augmented Generation (RAG) ensures accurate responses from curriculum data.

1.6 Assumptions

1.7 Limitation of the Study

- Limitations: The system may require periodic updates for improved interaction accuracy.

1.7.1 Technical Constraints

1. Hardware Limitations

- Computational Power

mini-PC must handle AI and LLM processing efficiently without excessive heat generation or power consumption.

- Battery Life

  Consider energy-efficient motors, sensors, and AI processing to maximize operational hours.

- Mechanical Complexity

  Ensure smooth and natural movement with a robust mobile base, including actuators and linkages in arms and head.

- Sensor Accuracy

  High-resolution cameras and microphones are needed for effective human interaction, but they must balance cost and power consumption.

- Weight Distribution

  A well-balanced design ensures stability when moving and interacting with humans.

2. Software and AI Limitations

- Real-Time Processing

  Delays in speech recognition or posture analysis can affect user experience. Optimize processing efficiency.

- LLM Integration

  Requires fine-tuning and fast response times while maintaining contextual understanding.

- Navigation and Mapping

  Implement SLAM and obstacle detection algorithms for safe and autonomous movement.

- Robustness in Unstructured Environments

  The AI must adapt to varying lighting, noise, and terrain conditions.

1.7.1.1 Non-Technical Design Constraints

- User Acceptance and Adaptability

  The robot must be designed for ease of use, ensuring that individuals of different technological backgrounds can interact with it effectively.

- Privacy and Data Security

  Handling user interactions requires adherence to data protection policies to maintain confidentiality and security.

- Cost and Scalability

  The system should be cost-effective for widespread adoption, balancing hardware affordability with advanced AI capabilities.

- Regulatory and Ethical Considerations

  Compliance with safety regulations and ethical AI deployment guidelines is essential to prevent bias, ensure transparency, and promote responsible AI usage.

1.8 Definitions

Abbreviation	Full Term	Definition
AI	Artificial Intelligence	Simulation of human intelligence processes by computer systems.
AMCL	Adaptive Monte Carlo Localization	Probabilistic localization algorithm for ROS-based autonomous navigation.
API	Application Programming Interface	Rules and protocols enabling software applications to communicate and exchange data.
ASR	Automatic Speech Recognition	Technology that converts spoken speech into written text.
LiDAR	Light Detection and Ranging	Remote sensing sensor that measures distances using laser pulses.
LLM	Large Language Model	AI model trained on large text datasets for understanding and generating human language.
NLP	Natural Language Processing	Subfield of AI concerned with enabling computers to understand human language.
RAG	Retrieval-Augmented Generation	Technique that enhances LLM outputs by retrieving relevant external information.
ROS	Robot Operating System	Open-source middleware framework for developing robot software.
SLAM	Simultaneous Localization and Mapping	Computational technique allowing robots to map unknown environments while tracking position.
STT	Speech-to-Text	Process of converting spoken audio signals into written text.
TTS	Text-to-Speech	Speech synthesis technology that converts written text into audible output.

CHAPTER 2

SYSTEM DESIGN AND RELATED TECHNOLOGIES

2.1 Mechanical Design

This section covers the mechanical aspects of the robotic assistant's design, including chassis, mobility systems, and structural components.

2.2 Electrical Design

This section covers the electrical systems, power management, and hardware integration aspects of the robot.

2.3 Software Design

2.3.1 Mobile Robot

2.3.1.1 ROS 2 (Robot Operating System 2)

ROS 2 is an open-source robotic middleware framework that has become the de facto standard for robot software development. Designed from the ground up to overcome limitations of the original ROS, it provides comprehensive support for modern robotic systems, including multi-robot teams, resource-constrained embedded platforms, real-time control systems, and operation across unreliable networks. ROS 2 establishes well-defined design patterns and architectural conventions for structuring complex robotic software, supporting modularity, scalability, and maintainability in autonomous systems. It targets production-grade deployment environments where reliability and determinism are critical, combining distributed middleware capabilities with tools and libraries that streamline development of sophisticated control, perception, and communication subsystems. For this project, ROS 2 serves as the core integration middleware connecting all robot subsystems.

2.3.1.2 ROS Navigation Stack

The ROS Navigation Stack provides a comprehensive framework for autonomous mobile robot navigation, accepting odometry data and sensor stream inputs and outputting velocity commands to the robot's motor controller. This creates a clean abstraction that separates high-level path planning logic from low-level motor actuation. The Navigation Stack requires correct configuration of several components: ROS middleware access, a tf transform tree defining coordinate frame relationships between hardware components, standardised sensor data publication using ROS message types, and tuned costmap and planner parameters reflecting the robot's geometry and dynamics. This modular architecture enables robust navigation in diverse environments while supporting independent testing and development of individual components.

2.3.1.3 AMCL (Adaptive Monte Carlo Localization)

Adaptive Monte Carlo Localization (AMCL) is the primary localization package used within the ROS ecosystem for autonomous navigation tasks. The algorithm employs a particle filter approach, maintaining a probabilistic distribution of candidate robot poses and iteratively refining this distribution as new sensor observations

and odometry measurements become available. AMCL is specifically optimised for use with 2D laser rangefinders and pre-built occupancy grid maps, making it well-suited to structured indoor environments.

It is important to note that AMCL is subject to specific operational constraints. The algorithm assumes a fixed geometric relationship between the laser sensor and the robot base; any movement of the laser relative to the base frame violates this assumption and can degrade localization accuracy significantly. Furthermore, AMCL requires continuous odometry input to propagate the particle filter over time. Odometry estimates may be derived from multiple complementary sensor sources, including wheel encoders and inertial measurement units (IMUs), which together provide more robust position and orientation estimates than either source alone.

2.3.2 Database Management

2.3.2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a technique that enhances large language model output quality by incorporating an external knowledge retrieval step before response generation. Rather than relying solely on patterns in the model's training parameters, RAG queries a domain-specific knowledge base at inference time, retrieving contextually relevant information to augment the model's context.

RAG offers cost-effective extension of LLM capabilities to specialised domains without expensive model retraining. By grounding responses in retrieved, current information, RAG substantially reduces factual hallucinations and improves relevance and accuracy. In this implementation, RAG forms the backbone of the knowledge retrieval subsystem, enabling contextually appropriate responses to user queries about the operating environment and domain-specific topics.

2.3.2.2 Vector Database

A vector database stores data as dense numerical vectors (embeddings) rather than conventional row-column structures, enabling similarity-based retrieval over large unstructured data collections including text, images, and audio. Vector databases support approximate nearest-neighbour search rather than exact-match queries, enabling retrieval of semantically related records even with differing query wording. This functionality is essential for RAG systems where semantic relevance rather than lexical overlap determines retrieval quality.

2.3.2.3 Embeddings

Embeddings are dense, fixed-dimensional numerical representations of objects — including text, images, and audio — produced by machine learning models trained to capture semantic and contextual relationships. Objects that are semantically similar are mapped to nearby points in the embedding space, enabling distance-based similarity measures to serve as a proxy for meaning. In this system, text embeddings represent both stored knowledge base documents and incoming user queries, enabling the vector database to identify and retrieve the most relevant passages for augmenting the LLM's response.

2.3.2.4 Milvus

Milvus is an open-source vector database platform specifically engineered for large-scale similarity search on massive embedding collections. Designed with production deployment in mind, Milvus provides highly optimised indexing structures and query algorithms that enable fast, accurate retrieval of semantically similar vectors from datasets containing millions or billions of embeddings. Unlike traditional relational databases, Milvus

is purpose-built for vector search operations, supporting multiple similarity metrics including Euclidean distance, cosine similarity, and inner product.

Key features of Milvus include exceptional scalability through distributed architectures that allow horizontal expansion, high-performance approximate nearest-neighbour search capabilities that retrieve relevant results with minimal latency even on very large datasets, and flexible indexing strategies optimised for different trade-offs between search accuracy and computational efficiency. In this implementation, Milvus serves as the underlying vector database infrastructure for the RAG pipeline, enabling rapid semantic retrieval of domain-relevant information from the knowledge base to augment LLM responses.

2.3.2.5 SQL (Structured Query Language)

SQL is a standardised declarative programming language for defining, manipulating, and retrieving data in relational database management systems (RDBMS). Data is organised in tables with rows representing records and columns representing typed attributes. SQL provides comprehensive operations for schema creation, record insertion and updates, referential integrity enforcement, and complex multi-table queries through joins and aggregations. Query optimisation and index management facilities ensure retrieval performance at scale. In this system, SQL maintains structured records of user interactions, profiles, and system logs, complementing the vector database layer for semantic retrieval.

2.3.3 Artificial Intelligence

2.3.3.1 Face Recognition

Facial recognition is a biometric identification technology that authenticates or identifies individuals by analysing distinctive geometric and photometric features of the human face. It detects facial regions, generates compact feature representations, and compares them against stored facial templates. The system automatically identifies returning users and personalises interactions based on stored preferences and interaction history. This capability enhances human-robot interaction naturalness and continuity by eliminating explicit identification steps. For the robotic assistant, facial recognition enables context-aware, personalized engagement with individual users.

2.3.3.2 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a multidisciplinary field enabling computational systems to understand, interpret, and generate human language in written and spoken forms. It combines rule-based linguistics, statistical models, and deep learning architectures for tasks including parsing, sentiment analysis, intent detection, and machine translation. NLP systems capture not only surface language form but also deeper semantic content, including speaker intent, implied meaning, and emotional tone. This nuanced understanding enables the robotic assistant to engage in natural, contextually appropriate dialogue, especially in open-domain settings with ambiguous or colloquial queries.

2.3.3.3 Large Language Model (LLM)

A Large Language Model (LLM) is a class of artificial intelligence system trained on massive corpora of text data — often comprising hundreds of billions of tokens — to acquire broad and deep competencies in understanding and generating human language. LLMs are built on the transformer neural network architecture,

which leverages self-attention mechanisms to capture long-range dependencies between tokens and model the statistical structure of language at multiple levels of abstraction.

Through large-scale pre-training followed by task-specific fine-tuning or instruction alignment, LLMs develop the capacity to perform a diverse array of language tasks — including question answering, summarization, translation, and code generation — without requiring explicit task-specific training data for each application. In this system, the LLM serves as the central reasoning and response generation component of the conversational AI subsystem, processing retrieved contextual information from the RAG pipeline to produce coherent, factually grounded responses to user queries.

2.3.3.4 Text-to-Speech (TTS)

Text-to-Speech (TTS) is a speech synthesis technology that converts written text into natural-sounding spoken audio, enabling AI systems to communicate verbally with users without requiring pre-recorded audio assets. Modern TTS systems are built on deep learning architectures — including neural vocoders and sequence-to-sequence models — that generate speech with dynamic, context-sensitive variations in prosody, pitch, speaking rate, and pronunciation. These characteristics produce output that closely approximates natural human speech patterns, significantly improving comprehensibility and user engagement compared with earlier rule-based synthesis methods.

TTS technology has found widespread application in accessibility tools for visually impaired users, virtual assistants, and interactive voice response systems. For the robotic assistant, TTS serves as the primary output modality for verbal communication, enabling the robot to deliver responses audibly in environments where visual screen interaction is impractical or undesirable.

2.3.3.5 Automatic Speech Recognition (ASR)

Automatic Speech Recognition (ASR) is a technology that processes raw acoustic speech signals and converts them into machine-readable text, forming the primary input pathway for voice-driven human-machine interaction. Contemporary ASR systems employ end-to-end deep learning architectures — including recurrent neural networks and transformer-based models — trained on large-scale speech corpora to achieve high transcription accuracy across a range of speakers, accents, and acoustic conditions.

Advanced ASR implementations integrate Natural Language Processing modules downstream of the transcription stage, enabling the system to infer the user's intent and semantic meaning from the recognised text in addition to producing a verbatim transcript. ASR accuracy is sensitive to several environmental factors, including background noise levels, microphone quality, and speaker proximity; consequently, robust noise handling and signal preprocessing are important design considerations for deployment in real-world robotic environments. Despite these challenges, modern ASR systems are capable of achieving near real-time transcription latency, enabling fluid conversational interactions between the user and the robotic assistant.

CHAPTER 3

RESEARCH METHODOLOGY AND SYSTEM IMPLEMENTATION

The students must study the requirements and formatting for their theses according to the thesis guidelines of the graduate studies in order to ensure that the paper outline meets the standard. Prior to the final thesis defense presentation, students are required to submit the original version to their thesis advisor for review to ensure accuracy and uniformity.

3.1 PAPER SIZE

The paper used for printing the thesis content must be white, unlined, A4 size paper (210 mm wide and 297 mm long).

3.2 FONT

For the thesis cover, use 20-point font size for the title in Thai (approximately 3 mm in height) and 18-point font size for the title in English (approximately 2.5 mm in height). Use the TH Sarabun New font style. The font size for the inside cover should be the same as the outside cover.

The font size for the chapter titles, section headings, and content can be found in the following section.

3.3 Software Design & Development

Use a 7-character word space, which is approximately 60 characters per line. Please indent the first line of each paragraph.

3.3.1 Software Architecture Overview

The top margin should be 1 inch (25.4 mm) from the paper's edge.

The left margin should be 1.5 inches (38.1 mm) from the paper's edge.

The right margin should be 1 inch (25.4 mm) from the paper's edge.

The bottom margin should be 1 inch (25.4 mm) from the paper's edge.

3.3.2 Database Design

The robotic assistant system employs a hybrid database architecture combining vector and relational storage to optimize both semantic retrieval and structured data management. This two-tier approach leverages the complementary strengths of each database technology to support rapid knowledge retrieval for RAG operations while maintaining referential integrity for transactional data.

3.3.2.1 Milvus Vector Database

Milvus serves as the primary storage layer for unstructured knowledge data including university information documents, curriculum materials, and timetable scheduling data. Rather than storing raw PDF files, images, and CSV documents directly, the system converts this content into dense vector embeddings through

encoding pipelines. This embedding-based representation enables semantic similarity searching, which substantially improves retrieval speed and relevance compared to traditional full-text search approaches.

The Milvus vector database is organized into three primary collections:

- **uni_info**: Stores university information including building layouts, facility locations, administrative details, and general campus information. These documents are converted to embeddings (512-dimensional) for efficient semantic retrieval.
- **curriculum**: Maintains curriculum-related documents, course descriptions, learning objectives, and academic program information. The embeddings allow RAG to quickly retrieve relevant curriculum content when students query course details.
- **time_table**: Manages timetable and scheduling information, storing class schedules, room assignments, and temporal constraints as vectors for fast retrieval of scheduling queries.

When the LLM processes user queries within the RAG pipeline, Milvus enables rapid approximate nearest-neighbor search to identify semantically similar documents, significantly reducing response latency compared to searching raw, unstructured data.

3.3.2.2 MySQL Relational Database

MySQL maintains structured relational data supporting the system's transactional and operational requirements. The relational schema enforces data integrity through foreign key constraints and normalization, ensuring consistency across user sessions and system operations. MySQL stores the following primary tables:

- **Academic_Year**: Records academic calendar information, including semester start/end dates and any relevant administrative periods linked to the system's operational timeline.
- **Students**: Maintains student enrollment records containing student identifiers, personal information (first name, last name, nickname), contact details (email), and enrollment year, serving as the authoritative registry for user identification and authentication.
- **Face_Recognition_Data**: Stores facial biometric templates and encodings along with associated student identifiers, enabling the face recognition subsystem to authenticate returning users and personalize interactions.
- **ExcelTimetableData**: Maintains structured timetable information including class schedules and room assignments, providing queryable access to scheduling constraints and student coursework.

3.3.2.3 Database Synchronization

The two databases maintain coordinated state through carefully designed synchronization procedures. Timetable data, in particular, serves both systems: the ExcelTimetableData table in MySQL stores structured scheduling information, while synchronized embeddings in the Milvus time_table collection enable semantic retrieval of scheduling information during conversation. When data mismatches or updates occur, both storage layers are updated together to maintain consistency across the system.

3.3.3 LLM Integration Module

The LLM Integration Module serves as the core conversational intelligence engine, implementing a five-path Retrieval-Augmented Generation (RAG) router that determines the optimal data source for each user query. This module employs a dual-model LLM architecture: a 70-billion-parameter primary model (Llama 3.1 Typhoon2-70B-Instruct) for high-quality conversation synthesis, and an 8-billion-parameter fast model (Llama 3.1 Typhoon2-8B-Instruct) for lightweight preprocessing tasks such as grammar correction and memory summarization. Both models execute via Ollama inference server on NVIDIA CUDA for low-latency operation.

3.3.3.1 Query Routing Strategy

The chatbot employs keyword-based intent detection to route queries to the most relevant data source. Routing is performed via exact-match keyword lookup against six possible destinations:

- **Conversation History**: Triggered by meta-conversational keywords (e.g., “ครั้งที่แล้ว”, “เมื่อกี้”, “ก่อนหน้า”, “ถามอะไร”, “history”, “previous”). Retrieves summary embeddings from Milvus conversation_memory collection using **BAAI/bge-m3** 1024-dimensional embeddings, enabling semantic memory of previous student interactions across sessions.

- **Student Information**: Queries about student rosters, names, contact information, or enrollment data (e.g., “นักศึกษา”, “ชื่อ”, “อีเมล”, “นศ”, “student”). Executes direct SQL queries against MySQL **Students** and **Academic_Year** tables, returning structured roster information.

- **Timetable & Scheduling**: Queries regarding class schedules, exam times, or room assignments (e.g., “ตารางเรียน”, “คาบเรียน”, “เวลาเรียน”, “วันไหน”, “เวลาไหน”, “กี่โมง”, “schedule”, “exam”). Retrieves semantically similar entries from Milvus **time_table** collection. Thai-to-English keyword augmentation automatically appends English synonyms (e.g., “คอร์ส” to “course”, “สอน” to “teach”) before embedding to improve multilingual matching.

- **Curriculum Information**: Queries about courses, credit hours, subject content, or academic programs (e.g., “วิชา”, “หลักสูตร”, “หน่วยกิต”, “เนื้อหา”, “course”, “subject”, “credit”). Performs semantic search over the Milvus **curriculum** collection (516 entities at 1024-dim).

- **University Information**: Queries about campus facilities, building locations, or zones (e.g., “อยู่ที่ไหน”, “ตึก”, “อาคาร”, “แผนที่”, “โซน”, “zone”). Default fallback route retrieving from Milvus **uni_info** collection (52 buildings across zones A–D).

- **General Chat**: Fallback for unroutable queries; uses conversation history and general knowledge without external data source.

3.3.3.2 Personalized Prompt Generation

The chatbot dynamically constructs system prompts that personalize each response based on student demographics, academic year, and context-aware state. Year-specific tone masks guide the LLM’s communication style:

- **Year 1 (Freshmen)**: “โทนการพูด: ให้กำลังใจ อบอุ่น เป็นกันเอง เหมือนรุ่นพี่คอยดูแลน้องใหม่” — Mentoring, supportive tone.
- **Year 2 (Sophomores)**: “โทนการพูด: สนับสนุนด้านโปรเจกต์และการเรียน สนใจความคืบหน้า” — Project-focused inquiry.
- **Year 3 (Juniors)**: “โทนการพูด: เน้นเรื่องฝึกงานและประสบการณ์ทำงาน ให้กำลังใจในช่วงนี้” — Internship and practical experience support.
- **Year 4 (Seniors)**: “โทนการพูด: เน้นโปรเจกต์จบการศึกษา ให้กำลังใจและแสดงความเชื่อมั่น” — Thesis support with confidence-building language.

Each prompt injection includes: student display name (from **thai_name** or cleaned **person_id**), academic year (auto-calculated from **enrollment_year**), year-specific tone, current date/time with Thai day-of-week abbreviation, retrieved conversation summary (1–2 sentences), and semantically retrieved RAG context (top-k=3 results). Explicit constraints enforce: Thai-only responses, prohibition of masculine particles (“ครับ”, “ผม”) in favor of feminine (“ค่ะ”, “หนู”), single-sentence response length for TTS readability, and robot self-location awareness (“น้องสาธุทำงานที่ชั้น 12 อาคาร E-12 โซน D”).

3.3.3.3 Output Schema & Intent Classification

The LLM returns structured JSON containing response text and detected user intent. Intent classification determines downstream routing to TTS and optional navigation commands:

- **chat**: General conversational response; TTS playback only.
- **info**: Factual information delivery; TTS playback only.
- **navigate**: Student explicitly requests navigation to a location. Response text confirms destination; triggers parallel TTS output + ROS2 navigation dispatch.
- **farewell**: Student indicates end-of-interaction. Response text bids goodbye; triggers parallel TTS output + ROS2 “resume_roaming” command.

3.3.3.4 Session Memory & Persistence

After each successful detection event, the module asynchronously stores conversation metadata via fire-and-forget background tasks without blocking the response pipeline. Text storage proceeds on two tiers:

1. **SQLite Raw Log**: All user-bot exchanges (both registered and guest) are immediately appended to the **conversation_log** table with exact transcripts, timestamps, and detected intent. This log enables debugging, audit trails, and offline conversation analysis.
2. **Milvus Memory Embedding**: For registered students only, the 8B-parameter fast LLM generates a summary (1–2 Thai sentences) capturing the interaction’s essence (student query + bot response core content). This summary is embedded via BAAI/bge-m3 (1024-dim) and stored in the **conversation_memory** collection partitioned by **student_id**, enabling semantic memory retrieval in future sessions.

This dual-write architecture ensures both immediate queryable raw logs and long-term semantic recall while guest interactions remain logged for compliance without cluttering the memory index.

3.3.4 Speech-to-Text (STT) & Grammar Correction Module

The Speech-to-Text module operates in two phases: (1) raw transcription via Typhoon2-Audio sidecar service running on a separate Python 3.10 process, and (2) Thai grammar normalization via the 8B-parameter fast LLM. This two-stage architecture separates acoustic modeling (sidecar responsibility) from linguistic polish (main server responsibility), minimizing latency impact on the core conversation pipeline.

3.3.4.1 ASR via Typhoon2-Audio Sidecar

The Typhoon2-Audio service (**typhoon_audio.py**) runs on FastAPI port 8001, exposing a **POST /stt** endpoint that accepts raw WAV bytes (16-bit PCM, 16 kHz sampling) and returns a JSON response containing **text: str** with the Thai transcription. The sidecar loads the **typhoon-ai/llama3.1-typhoon2-audio-8b-instruct** model with the parameter **attn_implementation="eager"** to avoid GLIBC incompatibilities on Ubuntu 20.04. Transcription latency is typically 1–3 seconds per 10-second audio clip. The main server communicates with the sidecar via HTTP, allowing transparent model swapping or sidecar reboots without main process interruption.

3.3.4.2 Thai Grammar Normalization

Raw ASR output frequently contains spelling errors, phonetic mishearings, and diacritical mark inconsistencies. The Grammar Corrector module (refactored as "STT Text Normalizer") uses the 8B-parameter fast LLM with aggressive parameter constraints to refine transcriptions into standard Thai orthography:

- **Message Input**: Raw ASR text (e.g., "อยากรู้ตารางเรียนนะ" or "ห้องไหน")
- **System Prompt**: Instructs the model to correct Thai spelling and phonetic errors while preserving emphasis particles ("ค่ะ", "นะ", "ครับ"), word order, and original meaning. Explicitly bans elaboration or reinterpretation.
- **Parameters**: **temperature=0.1** (minimal variation for consistent corrections), **max_tokens=256** (length-bounded).

The model is instructed: "ห้ามเพิ่มข้อมูล ห้ามตอบคำถาม ห้ามตัดทอนประโยค" — corrections only, no enrichment.

3.3.4.3 Input Filtering & Length Guards

To reduce unnecessary LLM calls and protect against edge cases:

- **Minimum Input Length**: Inputs shorter than 15 characters (typically noise or incomplete speech) are skipped; the raw transcription passes through unmodified.
- **Output Length Guard — Lower**: If corrected text is less than 50% of the input length, the output is discarded as a likely hallucination. The original raw text is retained instead. Example: input 50 characters to output 20 characters to reject, use raw.

- **Output Length Guard — Upper**: If corrected text exceeds 150% of the input length, it is similarly discarded as over-elaboration. Only corrections within the 50–150% band are accepted.
- **LLM Failure Fallback**: On timeout (15-second limit for 8B model) or inference server unavailability, the raw transcription passes through unchanged; pipeline does not halt.

These guards ensure robustness in production while maintaining the principle that STT correction is normalization, not augmentation.

3.3.4.4 Monitor Logging

Grammar-corrected text is logged to the '/monitor' dashboard as the "STT" row, distinct from the raw transcription. Empty or noise-only inputs display as "(empty)" in grey to visually distinguish from legitimate short utterances (e.g., "ได้", "ไม่")

3.3.5 Text-to-Speech (TTS) Module

The Text-to-Speech subsystem converts natural language responses into audible Thai speech via a multi-stage preprocessing and synthesis pipeline. The module must handle Thai phonetic complexity (tone marks, vowel placement, syllable boundaries) while expanding non-Thai textual elements (English letters, acronyms, numbers) into readable Thai forms. A configurable architecture allows runtime selection between server-side and edge synthesis options.

3.3.5.1 Text Normalization & English Number Expansion

Before synthesis, text undergoes four sequential normalization stages:

1. **Expansion of English Letters & Acronyms**: English words and acronyms are expanded to Thai spell-out form. Examples: - "E-12 ตึก" to "อี สิบสอง ตึก" - "KMITL" to "เค เอ็ม ไอ ที แอล" - "Zone D" to "โซน ดี"
2. **Number-to-Thai Translation**: Numbers are converted to Thai words. Four-digit room codes (e.g., "1201") are read digit-by-digit (หนึ่ง สอง ศูนย์ หนึ่ง); year numbers and single-digit room specifiers are read naturally.
3. **Unicode Normalization**: Apply NFC (Canonical Decomposition + Recomposition) via pythainlp to ensure consistent representation of Thai diacritical marks (tone marks, vowels, below-mark characters).
4. **Mai Yamok Expansion**: The Thai repetition mark "ๆ" is expanded to the preceding word using regex matching on Thai character ranges, preventing synthesis models from misinterpreting this control character.
5. **Syllable Tokenization**: Thai text is segmented using **pythainlp.word_tokenize()** for word-level boundaries, then each word is further decomposed into syllables via the **thai_syllables** library. Syllable boundaries are marked with spaces, preserving all diacritical marks in place. Non-Thai content (small English fragments) is preserved unchanged between Thai syllables.
6. **Whitespace Collapse**: Multiple consecutive spaces are reduced to single spaces, producing the final synthesis-ready form.

Example transformations:

- "คณะวิศวกรรมหุ่นยนต์" to "คณะ วิศ วะ การ หุ่น ยนต์"
- "ตึก E-12 ห้อง 1201" to "ตึก อี สิบสอง ห้อง หนึ่ง สอง ศูนย์ หนึ่ง"

3.3.5.2 Synthesis Engine Selection

The system supports two architectural variants configurable via **config/settings.yaml** without code modification:

Option A — Server-Side Synthesis (Active): Text is transmitted to the Typhoon2-Audio sidecar (**POST /tts** on port 8001), which synthesizes via the Typhoon-Audio 8B model into PCM WAV (16 kHz, 16-bit). Typical latency: 2–4 seconds per sentence. The server then POSTs the resulting WAV bytes to the Raspberry Pi 5 **/audio_play** endpoint for playback. Configuration: **tts_engine: typhoon audio**.

Option B — PI 5 Edge Synthesis (Legacy): Syllable-spaced text is transmitted to the Raspberry Pi 5 **/tts_render** endpoint, which performs synthesis locally on ARM CPU. This approach reduces server load but incurs ARM inference latency (3–8 seconds typical). Configuration swap: change **tts_engine** to **khanomtan** in YAML.

The Typhoon2-Audio sidecar is the preferred engine: combined network time + GPU inference (2–4s) is typically faster than ARM synthesis alone.

3.3.5.3 Fire-and-Forget Parallel Dispatch

To minimize response latency blocking, the Intent Router fires TTS and navigation commands asynchronously via **asyncio.create_task()** (fire-and-forget) rather than **await**. This decouples the main detection pipeline from PI 5 response times:

- **chat, info**: TTS playback is spawned; pipeline immediately returns success to PI 5.
- **navigate**: TTS audio confirmation (e.g., “ได้เลยค่ะ ตามหนูมาเลยนะคะ หนูจะพาไปที่ห้องสมุดค่ะ”) and ROS2 navigation POST with JSON containing **cmd: go to, destination: ห้องสมุด** are spawned in parallel; both run in background. No destination is spoken a second time (the LLM reply already names it).
- **farewell**: TTS audio (e.g., “ลาก่อนค่ะ”) and ROS2 resume command with JSON containing **cmd: resume roaming** are spawned in parallel.

This pattern eliminates the 15–25 second blocking observed when TTS was awaited synchronously on slow PI 5 networks, freeing the server to accept the next student interaction immediately.

1.1 Major Heading . . . . . .

1.1.1 Sub heading of 1.1 . . . . . .

1.1.1.1 Sub Heading of 1.1.1 . . . . . .

1) Sub Heading of 1.1.1.1 . . . . . .

For any section headings, capitalize the first letter of each word. The first letter of each word in major headings and subheadings must be capitalized.

Use black Thai Sarabun New font style with a 16-point font size (approximately 2 mm in height) for the content text. Maintain the same font style throughout the document for symbols or characters that the printer does not provide. Use black ink, preferably of high quality.

3.4 TABLES

Insert tables within the relevant sections of the main text, leaving one line space before printing the word ”Table” followed by the table number in bold, aligned to the left margin. Then, print the table name. If the table name is longer than one line, print the top line longer than the bottom line, starting in line with the first letter of the table name. Continue the table without leaving a line if needed. If a table is too wide to fit on one page and must be continued on the next page, align it to the left margin and print the word ”(continued)” followed by the table number, like ”Table 3.1”. After finishing the table, leave one line space before the next regular text.

(Leave one line space)

Table 3.1 Example of Table

Spacing S_1 (cm)	Spacing S_2 (cm)	Modulus %				
		0	30	50	80	100
3.3	7.5	49 cm²	64 cm²	81 cm²	85 cm²	90 cm²
3.8	8.5	60 cm²	75 cm²	90 cm²	97 cm²	100 cm²
4.3	9.5	70 cm²	81 cm²	95 cm²	99 cm²	109 cm²
4.8	10.5	–	–	–	–	–

(Leave one line space)

3.5 IMAGES

To insert images, leave one line space before placing the image in the center of the page. Include the words ”Picture.” or ”Figure” (use either one consistently throughout the document), followed by the image number, using bold font. The image description should not exceed one line and should be placed in the center beneath the image. If the description extends beyond one line, place it on the left edge. Number the images or pictures in a manner similar to how tables are numbered, following the end of the image description. Leave one line space before continuing with regular text.

(Leave one line space)

Figure 3.1 Example of Figure

(Leave one line space)

3.6 TABLE OF CONTENTS, TABLE OF TABLES, TABLE OF FIGURES (OR LIST OF FIGURES)

Print the words ”Table of Contents,” ”Table of Tables,” ”Table of Figures” (or ”List of Figures”) in the center of the page, 1 inch from the top margin, using 24-point bold font.

Leave one line space and then print the word ”Page” aligned to the right. Print the page numbers flush with the right margin and connect them to the content with dots.

Next, after one line space, will be the content of the table of contents, between different chapters, bibliography, and appendices. Leave one line space. As for the Table of Tables and Table of Figures (or List of Figures), print the words ”Table #” or ”Figure #” aligned to the left margin, on the same line as ”Page.”

3.7 ABBREVIATION LIST OR SYMBOL LIST

Use this for cases when the abbreviation list is printed separately from the introduction and follows the instructions in section 4.10. Print the words ”Abbreviation List” or ”Symbol List” (or ”Abbreviation and Symbol List”) in the center of the page, 1 inch from the top margin, using 24-point bold font.

Leave one line space, and then start printing the abbreviations or symbols flush with the left margin. If the description of the abbreviation or symbol doesn’t fit on one line, start the next line from the 8th character, as before.

For English letters, Greek letters, and symbols with upper/lower marks and numeric marks: English Letters: A, B, C, b, c Greek Letters: C, P, Y,...

If there are English letters mixed with Greek letters, the Greek letters come after the English letters. Symbols with Upper/Lower Marks and Numeric Marks: A_1 A_x A^1 A^x A_1^x a_1 a_x a^1 a^x a_1^x

In this order, symbols with lower marks come before symbols with upper marks. Additionally, symbols with numeric marks come before symbols with alphabetic marks.

3.8 SCIENTIFIC NAMES

For biological organisms, plants, and animals, use the International Code of Nomenclature for emphasizing scientific names. Make them stand out distinctly from other text by underlining or using italics. Scientific names follow the binomial system, consisting of two parts. The first part is the Genus name, which begins with an uppercase letter, and the second part is the Specific epithet, written slightly offset and beginning with a lowercase letter. In scientific names, it’s common to include the name of the person who initially defined the name and described the organism. Personal names are typically used for surnames only. If it’s a well-known name, it may be abbreviated, such as Linnaeus abbreviated as ”Linn.” or ”L.” In some cases, when two individuals co-author a name, both names are used, for example...

a. micro-organism	Escherichia coli,
	Bacillus subtilis,
	Azospirillum brasilense
b. plant	Coccinia grandis L.,
	Canna indica Linn.,
	Cocos nucifera Linn.
c. animal	Ptilolaemus tickeli,
	Panthera tigris

3.9 MATHEMATICAL EQUATIONS

Mathematical equations can be inserted into the text. To maintain order, separate equations on their own lines. Equations should be positioned with a one-line space above and below. Equations should be centered on the page as appropriate and use the Cambria Math font with a font size of 11 points.

Enclose equation numbers in brackets, and order equation numbers in the same way as tables and images, for example...

$$S.D. = \sqrt{\frac{\sum(x-\bar{x})^2}{N-1}} \tag{3.1}$$

3.10 FOREIGN LANGUAGES

For words in foreign languages, print a translation in English within parentheses following the necessary terms. For example, Technique (เทคนิค). Printing foreign languages without diacritical marks is common. For instance, ”เทคโนโลยี” should be printed as ”Technology” Compound words should not use ”ส” or ”ส์” in Thai. For example, ”Games” should be printed as ”เกม,” except for proper names like ”SEAGAMES,” which should be printed as ”ซีเกมส์.”

3.11 ABBREVIATIONS

b&w (black and white) means black and white images.

c. (copyright) means used for copyright year.

ca. (circa) means an approximate date.

ch. (chapter) means chapters in royal decrees, laws, etc.

chap. (chapter) means regular chapters; use ”chaps.” for plural.

col. (color) means color photographs.

ed. (edition ; editor ; edited by) means editor or edited by.

enl. (enlarged) means an enlarged edition, e.g., ”enl. ed.”.

et al. (et alii) means and others.

fig. (figure) means figures, plural ”figs.”.

fr. (frame) means frame in slides to indicate the number of frames.

i.p.s. (inches per second) means inches per second in tape speed.

ill. (illustrated by) means illustrated by.

min. (minutes) means film duration in minutes.

ms. (manuscript) means manuscript, plural ”mss.”.

n.p. (no place; no place of publishing) means no place of publication.

no. (number) means number, plural ”nos.”.

2nd. ed. (second edition) means the second edition.

p. (page) means page, plural ”pp.”.

Par. (paragraph) means paragraph, plural ”pars.”.

Pt. (part) means part, plural ”pts.”.

r.p.m. (revolutions per minute) means revolutions per minute.

rev. (revised) means revised edition, e.g., ”rev. ed.”.

3rd ed. (third edition) means the third edition.

sc. (scene) means scene.

sd. (sound) means sound in films.

sec. (section) means section, plural ”secs.”.

Si. (silent) means silent films.

trans. (translator; translated by) means ”translator” or ”translated by”.

Vol. (volume) means volume, plural ”vols.”.

3.12 PUNCTUATION MARKS

Period (.) should be printed with a 2-space gap between words.

Comma (,) should be printed with a 1-space gap between words.

Semicolon (;) should be printed with a 1-space gap between words.

Colon (:) should be printed with a 1-space gap between words.

Quotation marks (” ”) should be printed with a 1-space gap between words.

CHAPTER 4

REFERENCE AND CITATION

Citation refers to the reference documents used in thesis writing. It is a way to inform the sources of the text, ideas, or any content that is not originally created by the author. This is done as a tribute to individuals or organizations who own the intellectual property of the ideas or the data. It also serves the convenience of readers who wish to know more details from the original source to be able to trace accurately. Citations can either summarize the original content or can be a direct quotation, maintaining the original writing format. To ensure accuracy, all information in the reference list must be precise and thoroughly checked for their sources.

Citations come in two main formats:

In-text citation using the name-year system.

In-text citation using the numerical sequence system.

At the end of the thesis, all the documents and references cited can be organized alphabetically according to the authors' names in a **bibliography** or in numerical order according to the reference numbers in a **reference** list.

4.1 IN-TEXT CITATION IN THE NAME-YEAR SYSTEM

Citations should include the author's name and the publication year, with the option to include page numbers separated by a colon (:).

Author names in Thai documents should include both first and last names, while in foreign language documents, only the last name should be used.

The citation format may vary depending on the sentence structure, either placed at the beginning or end of a sentence.

Example (Author mentioned at the beginning of the sentence):

Suttilak Ampanswong (2521: 25) explains that information refers to various knowledge, news, and data.

Good (1973: 112) defines commitment as an individual's expression of love and concern.

Example (Author mentioned at the end of the sentence):

...managers must have the ability to manage people to foster cooperation and collaboration in responsive and well-coordinated work, leading to the efficient achievement of organizational objectives (Uthai Boonprasert, 2531: 23).

In citations, there may be different cases as follows:

1. **Single Author**

Mayuree Chaisawat (2538: 86)...

Heyes (1964)...

2. **Two Authors**

Prayad Janthachomphu and Prasopas Aksonmat (2518: 24)...

Macaulay and Berkowitz (1978: 4)...

3. **More than Two Authors**

Sanan Jittrasuk et al. (2532: 21-25)...

Bradley, S. et al. (1983: 23-25)...

4. **Authors from an Institution, Organization, Corporation, or Agency**

From the meeting of education institution managers under the Department of Vocational Education (Department of Vocational Education, 2531) on the topic "Human Resource Needs."

...and environmental pollution, termed as 'air pollution' or the occurrence of air pollution (Department of Environmental Quality Promotion, 2539: 118).

5. **Multiple Titles by the Same Author in the Same Year**

Sutthas Yaksan (2529a)...

Heyes (1964c)...

6. **Multiple Citations within the Same Topic from Multiple Sources**

...(Yawonuch Sengyont, 2525a; Supada Intranukul, 2525)...

...(Kartner, 1973; Kartner and Russel, 1975)...

7. **No Author Identified**

Use 'Anonymous'

Anonymous (1973)...

8. **Quotations Longer Than 3 Lines**

Should be separated from the main text with a one-line space above and below. The left and right margins of both sides should be indented 8 character spaces, followed by the name-year citation in parentheses, without quotation marks (" ").

Example:

...The personnel in the organization are the heart of the organization, a vital force that propels the organization forward. As Somyasit Naweekarn stated,

(Leave 1 line blank)

'Internal conflicts within an organization can be designed or managed in various ways. Conflicts can yield positive and negative results. The positive aspect is that conflicts lead to the exploration of better outcomes, making the organization operate more efficiently. However, if conflicts are excessive, they can negatively impact the personnel in the organization and impede the achievement of the organization's goals' (Somyasit Naweekarn, 2530: 15).

9. **Electronic Information and Communication**

For electronic sources such as CD-ROM, Internet, use the previously mentioned citation methods.

4.2 IN-TEXT CITATION IN THE NUMERICAL SEQUENCE SYSTEM

In this referencing system, citations are numbered sequentially in the order they appear in the text, starting from the first chapter to the last. The reference numbers are enclosed in square brackets at the end of the relevant text. If a reference is repeated, the same number is used.

Example:

Due to the necessity of frequency-based decoding, various transformations must be applied, including essential conversions [1]. Apart from other computations, the prime factor algorithm is employed for

decoding, reducing the computational load when combined with the Short Length Algorithm [2]. In practical implementation, the application of these methods requires careful consideration...

Alternatively, Berlekamp's technique has the capability to generate both polynomials [2] . However, the process becomes more intricate in the presence of erasures. Blahut [3] has detailed the procedure for obtaining the error-erasure locator polynomial. This involves initializing the error locator polynomial with the erasure locator polynomial. He has also demonstrated that the error evaluator polynomial and the derivative of the error locator polynomial can be iterated within those 2t iterations [4].

CHAPTER 5

BIBLIOGRAPHY AND REFERENCES

The writing of a bibliography or references should adhere to one system that aligns with the chosen citation method. Use 'Bibliography' when citing in-text with the name-year system and use 'References' when citing in-text with the numerical sequence system. The use of bibliography management software, such as Endnote, Mendeley, Zotero, etc., is permitted and encouraged.

5.1 GUIDELINES FOR WRITING BIBLIOGRAPHY

1. Print the word Bibliography in the center of the page in bold, size 24 points.
2. Arrange the entire reference list together, alphabetically based on the first letter of each reference, following the alphabetical order as per the dictionary.
3. For theses written in Thai, list the bibliography in the Thai language first, followed by the English language bibliography. For theses written in English, list the English language bibliography first, followed by the Thai language bibliography.
4. Begin printing the bibliography at the left edge of the paper. If the entry does not fit on one line, start a new line indented by 7-character spaces. Begin printing the 8th character if the entry extends to the second line. If not completed within two lines, continue onto lines 3-4 starting from the same position as line 2 until the end of the entry. When starting a new entry, align it with the left edge of the paper without skipping a line.

5.2 GUIDELINES FOR WRITING REFERENCES

1. Print the word References in the center of the page in bold, size 24 points.
2. Arrange the reference documents in numerical order, according to the numbers assigned within parentheses referring to the content of the thesis.
3. There is no need to separate by language or document type.
4. Print the number of each entry at the left edge of the paper. If the entry does not fit on one line, start a new line indented by 7-character spaces. Begin printing the 8th character if the entry extends to the second line. When starting a new entry, align it with the left edge of the paper without skipping a line.

5.3 METHOD OF LISTING BIBLIOGRAPHY

5.3.1 Authors

1. The author's name should not include any titles such as Mr., Mrs., Miss, Dr., etc.
2. For entries in the Thai language, use the author's first name followed by the last name.
3. For entries in English, use the last name, followed by a comma and the abbreviated first and middle names in sequence.
4. If the author holds a doctorate (Ph.D.), professorship, or other title, include it after the name, separated by a comma.

5. Use the name as it appears in the document for authors with honorific titles.

6. If there are two authors, use the name of the first author followed by and between the names of the two authors.

7. If there are more than two authors, list all authors.

8. For authors using pseudonyms, use the pseudonym as it appears in the document.

9. For translated documents, list the name of the translator first, followed by the phrase "translated from" and the original author's name. If the original title is unknown, use the phrase "translated by" or "translated and compiled by."

10. For documents attributed to an organization, ministry, or department, use the name of the organization as the author. In the case of documents issued under a specific department within a ministry or subordinate to a department, even if the ministry's name is mentioned, use the department's name as the author. Include the name of the subordinate unit in the publisher section.

11. For documents with only an editor or compiler, use the editor's name followed by a comma and the word editor or compiler after the name of the author.

12. For documents with no stated author or conference proceedings, use the title of the document as the first entry.

5.3.2 Article Titles

1. Use the title as it appears in the document, enclosed in quotation marks.

2. For English titles, capitalize the initial letter of each word, except for conjunctions, articles, and prepositions, unless they are the first word.

3. If there is a scientific name in Latin for plants, animals, etc., insert it in *italics* or <u>underline</u> it.

5.3.3 Book or Journal Titles

1. Print book or journal titles in bold.

2. For international journals, you may use the correct abbreviation of the journal name.

3. For book series, if referencing a specific volume, include only that volume. For example, if referring to volume 3, and if citing multiple non-consecutive volumes, indicate each volume number with commas in between, such as Volume 1, 3, 5. If citing all volumes in the series, state the total number of volumes, such as 5 volumes or 5 vol. Underline or print in bold (either one) continuously from the title.

5.3.4 Editions

1. For the first edition or first printing, it is not necessary to specify in the bibliography.

2. Specify the edition number for subsequent editions, as indicated in the printed material. For example, 2nd Ed., 2nd printing, 2nd, rev. ed., or 2nd, expanded ed.

5.3.5 Place of Publication and Publisher

1. For Thai-language references, include the name of the province as the place of publication. For Bangkok, use the term Bangkok.

2. If the publisher has offices in multiple cities and the names of those cities appear in the document, use the first city mentioned as the place of publication.

3. List the publisher's name without including terms like Publisher, Company Limited, "Publisher," "Co—Inc.," or "Co..Ltd."

4. If the document does not mention the publisher's name, list the name of the printing house with the term Printing House.

5. For publishers associated with universities or organizations, provide the full name, such as Thammasat University Press.

6. If the printer is a government or private organization, use the organization's name as the publisher.

7. If the name of the person responsible for printing is not stated, use 'n.p.' (no place of publishing) in English.

5.3.6 Year of Publication

1. Include the year of publication in numeric format, as it appears in the document.
2. If the publication year is not mentioned, use 'n.d.' (no date) in English.

5.3.7 Example of Bibliography Format

The specifications for referencing in the bibliography format are as follows:

1. Variables in different listings, such as author, day, month, year, and publisher, should use the regular Courier PC format.
2. Variables that need to be printed in bold should use the Courier Upright Bold format.
3. Terms that need to be printed in italics, such as 'ผู้แปล' (translator) or 'โดย' (by), should use the regular Arial Unicode MS format.
4. Replace single-letter spacing with the '/' symbol.
5. In cases where a single line is not sufficient, start the second line with the eighth character, leaving a space of 7 characters.
6. Other symbols such as '.', ',', " ", :, -, [ ], should be printed as specified.

5.3.7.1 Book

Format:

Author.//Publication Year.//Title.//Volume (if applicable).//Edition (if applicable).//City of Publication.//Publisher.

Examples:

Single Author:

Sumner, M. 1990. Computer: Concept and uses. 2nd ed. New York: McGraw-Hill.

Two Authors:

Gilbert, A. and Gnglir, J. 1982.**Cities Poverty and Development Urbanization in the Third World.** London: Oxford University Press.

Multiple Authors:

Jackson, M.H. Stewart, D. and Steven, G. 1991.**Environmental Health Reference Book.** Oxford: Butterworth Heineman.

Authors with Pseudonyms:

น.ม.ส. (นามแฝง). 2496. นิทานของ น.ม.ส. พิมพ์ครั้งที่ 3. พระนคร: คลังวิทยา.

5.3.7.2 Translated Books

Format

Author.//Year BE. Publication Year.//**Book Title.**//Volume (if any).//Translated by Translator.//Edition (if any).//City of Publication://Publisher.

Or

Author.//Year BE. Publication Year.//Book Title.//Volume (if any).//Translated from**Original Title**.//by Translator.//Edition (if any).//City of Publication://Publisher.

Example

Davenport, J.H. 1993.**Computer Algebra.** Translated by Davenport, A and Davenport, J.H. 2nd ed. Great Britain : Academic Press.

Shafarevich, I.R. 1994. Basic Algebraic Geometry 1. translated from **Osnovy Algebraicheskoj Geometrii, tom 1.** by Reid, L. 2nd ed. New York : Springer.

5.3.7.3 Translated Books with Unspecified Original Author in the Original Language

Format

Translator Name,/Translator.//Year BE. Publication Year.//Book Title.//Volume (if any).//Edition (if any).//City of Publication://Publisher.

Example

Duang Bunnak, Translator. 2511. Economics. 2nd ed. Bangkok: Thammasat University.

Apichat Thaowto and Serm Sakkulkul, Translators. 2526. Basic Knowledge of Rice Cultivation. Bangkok: Kasetsart University.

5.3.7.4 Books Authored by Organizations

Format

Organization Name.//Year BE. Publication Year.//Book Title.//Volume (if any).//Edition (if any).//City of Publication://Publisher.

Example

The Bank of Thailand. 1992. 50 Years of the Bank of Thailand 1942-1992. Bangkok : Amarin Printing Group.

5.3.7.5 Editors or Compilers

Format

Editor Name,/Editor (Compiler).//Year BE. Publication Year.//Book Title.//Volume (if any).//Edition (if any).//City of Publication://Publisher.

Example

Friedman, E.G., editor. 1995. Clock Distribution Networks in VLSI Circuits and Systems. New York : IEEE Press.

5.3.7.6 Conference Proceeding or Articles from Books

Format

Author of the Article.//Year BE. Publication Year.// ”Article Title (Conference Report).”//Page/Page Range.//In//Editor (Compiler).//Book Title.//Volume (if any).//Edition (if any).//City of Publication://Publisher.

Example

Niwa, E. 1992. ”Chemistry of surimi gellation.” 115-118. in Lainier, T.C. and Lee, C.M. Surimi Technology. New York : Marcel Dekker

Goemans, M.X. and Williamson, D.P. 1995. ”The Primal-dual Method for Approximation Algorithms and its Application to Network design Problems.” 69-87. in Hochbaum, D. Approximation Algorithms for NP-hard Problems. Boston : PWS.

5.3.7.7 Journal Articles

Format

Author.//Publication Year.// ”Article Title.”//Journal Name.//Volume(Issue)/:/Page Range.

Bentley, M Lee. 1986. ”The Role of Backcountry Experience in Middle School Environmental Education,” Dissertation Abstracts Internation. 46(10) : 2900-A-2901-A.

Choomchuay, S. 1994. ”On the Implementation of Finite Field Operation.” Ladkrabang Engineering Journal. 11(1) : 7-16.

Hitchcock, R.B. et. al. 1982. ”Timing Analysis of Computer Hardware.” IBM J. Res. Develop. 26(1) : 100-105.

Klein, P. et. al. 1995. ”An approximate max-flow min-cut relation for undirected multi-Commodity flow, with applications.” Combinatorica. 15(2) : 187-202.

5.3.7.8 Thesis and Dissertation

Format

Author.//Year BE. Publication Year.// ”Thesis Title.”//Degree Name (Full).

Example

Choomchuay, S. 1993. ”Algorithm and Architecture for Reed-Solomon Decoding.” Ph.D.Thesis of University of London and the Diploma of Membership of the Imperial College.

Burns, B. 1991. ”Performance Analysis and Optimization of Asynchronous Circuits.” Ph.D. Dissertation, California Institute of Technology.

5.3.7.9 Reviews

Format

Reviewer Name.//Review of//Title of the Reviewed Work.//by//Author Name.//In//Book Title (where the review appears).//Edition (if any).//Page Range.

Example

Chamnan Nakprasom. Review of the book Lai Suea Siam by S. Siwarak. Social Perspectives Journal, Volume 5(1), Issue 1 (June-August 2510): 139-141.

5.3.7.10 Articles in Encyclopedias or Dictionaries

Format

Author.//"Article Title."//Encyclopedia or Dictionary Name.//Volume (Publication Year)/://Page Range.

Example

Prapat Trinanong. 2513. "Chor Fa." Thai Encyclopedia by the Royal Institute, Volume 10: 6076-6080.

5.3.7.11 Newspaper Articles

Format

Author.//Publication Year, Date, Month.//"Article Title."//Author of the Article.//Newspaper Name.//Page/Page Range.

Example

Goleman, D. 1985, 21 May. "New Focus on Multiple Personality." New York Times. Pages c1-c6.

5.3.7.12 News Monitoring and Facts Checking in Newspapers Over a Period

Format

Newspaper Name.//Date, Month//Year of Publication - Date, Month//Year of Publication.

Example

Thai Rath. January 4, 2538 - February 5, 2538.

5.3.7.13 Newsletters, Photocopies, and Unpublished Documents

Format

Author.//Year BE. Publication Year.//"Document Title."//Edition (if any).//City of Publication://Publisher.//Document Copy.

Example

Sakallah, K.A. et. al. 1989. "Analysis and Design of Latch-Controlled Synchronous Digital Circuits." Ann Arbor, Mi, : Dept. of EECS University of Michigan Technical Report CSE-TR-31-89.

5.3.7.14 Government Publications and Other Official Documents

Format

Organization Name.//Publication Year.//Publication Title.//Edition (if any).//City of Publication://Publisher.

Example

Office of the Secretary-General of the House of Representatives. 2522. Reports of the 8th-9th Representative Assembly Regular Session Volume 3.Bangkok:Printing Office of the Cabinet.

5.3.7.15 Interviews

Format

Interviewee Name,//Interviewer.//Date, Month Year of the Interview.//Interviewer Name//Interview Title.//Interview Location.

Example

Manas Sangwarasilp Interviewed, April 18, 2541. Somsak Chumchuay, Interviewer. Interview on the Improvement Guidelines for Graduate Programs of the Institution. Graduate School, King Mongkut's Institute of Technology Ladkrabang.

5.3.7.16 Audiovisual Materials

1) Slides, Filmstrips, Videos, Tape Recordings, Compact Discs

Format

Creator.//Year BE. Production Year.//Title.//[Type of Medium].//Place of Production://Producer.

Example

Charoen Prokpun Ltd. 2525. Chicken Farming. [Video]. Bangkok: Charoen Prokpun Ltd.

Alan W.B. Compiler. 1984. Using the Oscilloscope. [Slide]. London : The Slide Centre.

RCA Records. 1984. Test Compact Disc ; Extensive Test for Audio Equipment. [Compact Disc]. Germany : RCA Ltd.

2) Maps

Format

Name [Map Type].//Publication Year.//Publication Title.//Publisher.

Example

Scotland [Photographic map]. Washington (DC) : National Geographic Society (US.)

5.3.7.17 Patents

Format

Patentee.//Invention Name.//Country of Patent.//Patent Number.//Date of Patent Grant.

Example

Kamder, A.D. Method and Apparatus for Constraining the Compaction of Components of a Circuit Layout. U.S patent no. 5636132, June 1997.

5.3.7.18 Electronic Media

Format

Author.//Publication Year.//Title.//[CD-ROM].//Place of Production://Producer.

or

Author.//Publication Year (or Retrieval Date).//Title.//[Online].//Available://Access Method and Data Location.

Examples

Daniel H., Anghileri. 1995. Secondary Mathematics and Special Education Needs. [CD-ROM]. New York: Casell.

Noam E.M. 1994 Telecommunication Policy Issue for the Next Century. [Online]. Available: Gopher://198.80.36.../global/telecom.txt.

Texas Instruments. 1998. Speech synthesis Processors. [Online]. Available: http://www.ti.com/sc/docs/msp/speech/dex.htm.

5.3.8 Example Formats for References Citation

5.3.8.1 Book

Format

Author.//Book Title.//Edition (if any).//City of Publication://Publisher.//Publication Year BE.

or

Editor Name./Editor (Compiler).//Book Title.//Edition (if any).//City of Publication://Publisher.//Publication Year BE.

Example

[1] Richard E. Blahut. Theory and Practice of Error Control Codes. Reading : Addition-Wesley publishing Company, Inc.1984.

[2] Judith S.Van ALSTYNE. Professional and Technical Writing Strategies. International Edition. New JERSEY : Prentice-Hall, Inc. 1986.

[3] Jose E. Franca, Yannis Tsividis, Editors. Design of Analog-Digital VLSI Circuits for Telecommunication and Signal Processing. 2nd ED. New Jersey : Prentice-Hall, Inc.1994

[4] Microsoft. Hardware Design Guide for Window 95 ; A Practical Guide for Developing Plug and Play PCs and Peripherals. Washington D.C. : Microsoft Press. 1994.

[5] Microelectronics Group, Lucent Technology. Field-Programmable Gate Arrays [data Book]. LUCENT Technology. October 1996. Pp.2-36-2-40.

5.3.8.2 Articles from Journals, Books, or Conference Proceedings

Format

Author.//"Article Title."//Journal Name.//Volume,/Issue Number.//Month Year of Printing.//Page/Page Range.

Example

[1] Choomchuay S. "On the Implementation of Finite Field Operations" Ladkrabang Engineering Journal, vol.ll,no, l, June 1984. pp.7-17.

[2] Shao M., Reed S. "On the VLSI Design of a Pipeline Reed-Solomon Decoder Using Systolic Arrays." IEEE Trans. On Comput., vol. C-37, no. 10, Oct. 1988. pp. 1273-1280

[3] Sukiyama Y., Kasahara M., Hirasawa S. and Namekawa T. "A Method for Solving key Equation for Decoding Coppa Codes.", Inf. And Cont., vol.27, 1975. Pp. 87-99

[4] Choomchuay S., Arambepola B. "An Algorithm and a VLSI Architecture for Reed – Solomon Decoding" Proc. IEEE-ISCAS, San Diego, USA, May, 1992. Pp.2120-2123

5.3.8.3 Thesis and Dissertation

Format

Author.// ”Thesis Title.”// ”Full Degree Name.”//Year of Publication BE.

Example

[1] Choomchuay S. ”Algorithm and Architecture for Reed-Solomon Decoding.” Ph.D. Thesis of University of London and the Diploma of Membership of the Imperial College. 1993

5.3.8.4 Patents

Format

Patent Holder.//Invention Title.//Country of Patent.//Patent Number.//Date of Patent Registration (Day, Month, Year).

Example

[1] Kandar A.D. ”Method and Apparatus for Constraining the Compaction of Components of a Circuit Layout.” U.S patent no. 5636132, June 1997

5.3.8.5 Visual Materials and Other Media

Format

Producer.// ”Title.” //[Type of Media].//Place of Production.//Producer.//Year of Production BE.

Example

[1] RCA Records. ”Test Compact Disc ; Extensive Test Audio Equipment.” [Compact Disc]. Germany : RCA Ltd. 1984.

[2] Alan W.B. Compiler. ”Using the Oscilloscope.” [Slide]. London : The Slide Center. 1984.

5.3.8.6 Electronic Media

Format

Author.// ”Title.” //[CD-ROM].// Place of Production.//Producer.//Year of Release BE.

Or

Author.// ”Title.” //[Online].//Available at://://Access Method and Source.//Year of Publication BE (or retrieval).

Example

[1] Daniel H., Anghileri. ”Secondary Mathematics and Special Education Needs.” [CD-ROM]. New York : Casell.1995

[2] Noam E.M. ”Telecommunication Policy Issue for the Next Century.” [Online]. Available : gopher://198.80.36.../global/telcom.txt. 1994.

[3] Texas Instruments. ”Speech Synthesis Processors.” [Online]. Available : http://www.ti.com/sc/does/msp/speech/index.htm. 1998.

[1]

Bibliography

[1] Xuechen Zhang, Zhengmin Wu, Chengmao Cao, Kun Luo, Kuan Qin, Yangyang Huang, and Jie Cao. Design and operation of a deep-learning-based fresh tea-leaf sorting robot. *Computers and Electronics in Agriculture*, 206:107664, 2023.

Appendix A

GUIDELINES AND PROCEDURES FOR THESIS WRITING

Thesis writing is considered a crucial part for graduate students after completing coursework at a certain level. Students must plan their thesis writing process until the comprehensive thesis examination is completed.

Procedures for Requesting Approval of Thesis Topics and Scheduling Thesis Examinations:

1. Graduate students have the right to propose thesis topics as soon as they become enrolled in the graduate program.

2. Students must submit a request for approval of the thesis topic to the graduate school of their respective faculty.

3. The approval of the thesis topic and the thesis outline is within the authority of the dean, endorsed by the academic affairs committee.

4. Students must receive approval of the thesis topic and the outline (if any) before the final thesis examination.

5. Any changes before the thesis examination must be handled as follows:

5.1 **Change of thesis topic or outline:** Students must submit a request to the dean, with approval from the thesis advisor and endorsement from the academic affairs committee.

5.2 **Change of thesis advisor:** Students must submit a request to the dean of the respective academic unit, and the dean has the authority to approve, with endorsement from the academic affairs committee.

5.3 In cases where the thesis advisor loses their position within the academic unit, the student must find a new advisor, and the work specified by the owner of the work, including the student's name, the name of the thesis advisor, the name of the academic unit, and the name of the institution, should be used to support the thesis examination application.

6. The thesis examination is the final stage, and students must comply with the conditions and requirements of each program in their respective academic units. Students can only take the thesis examination when:

6.1 The thesis advisor believes that the student is ready for the examination. The student must notify the examination committee and submit a draft of the thesis to the specified number of graduate students, faculty, and the examination committee no less than two weeks in advance.

6.2 For master's degree examinations, the branch chairman and the course coordinator must conduct the thesis examination committee, endorsed by the academic affairs committee. The committee should consist of five members, including the thesis advisor, one graduate faculty member proposed by the thesis advisor, and three members, including an outside faculty member who is an expert, proposed by the academic affairs committee. The committee must appoint the external committee member as the committee chairman.

6.3 For doctoral degree examinations, the dean should appoint the thesis examination committee, endorsed by the academic affairs committee. The committee should consist of five members, including the thesis advisor, one faculty member proposed by the thesis advisor, and three members, including an outside faculty member who is an expert, proposed by the academic affairs committee. The committee must appoint the external

committee member as the committee chairman.

AUTHOR BIOGRAPHY

Name Mr. First Name Last Name

Date of Birth January 1, 1990, in Bangkok

John Doe was born in Bangkok, Thailand, on January 1, 1995. He received his Bachelor of Engineering degree in Robotics and AI Engineering from KMITL in 2017. He is currently pursuing a Master of Engineering degree in Robotics and AI Engineering at KMITL. His research interests include machine learning, computer vision, and robotics.